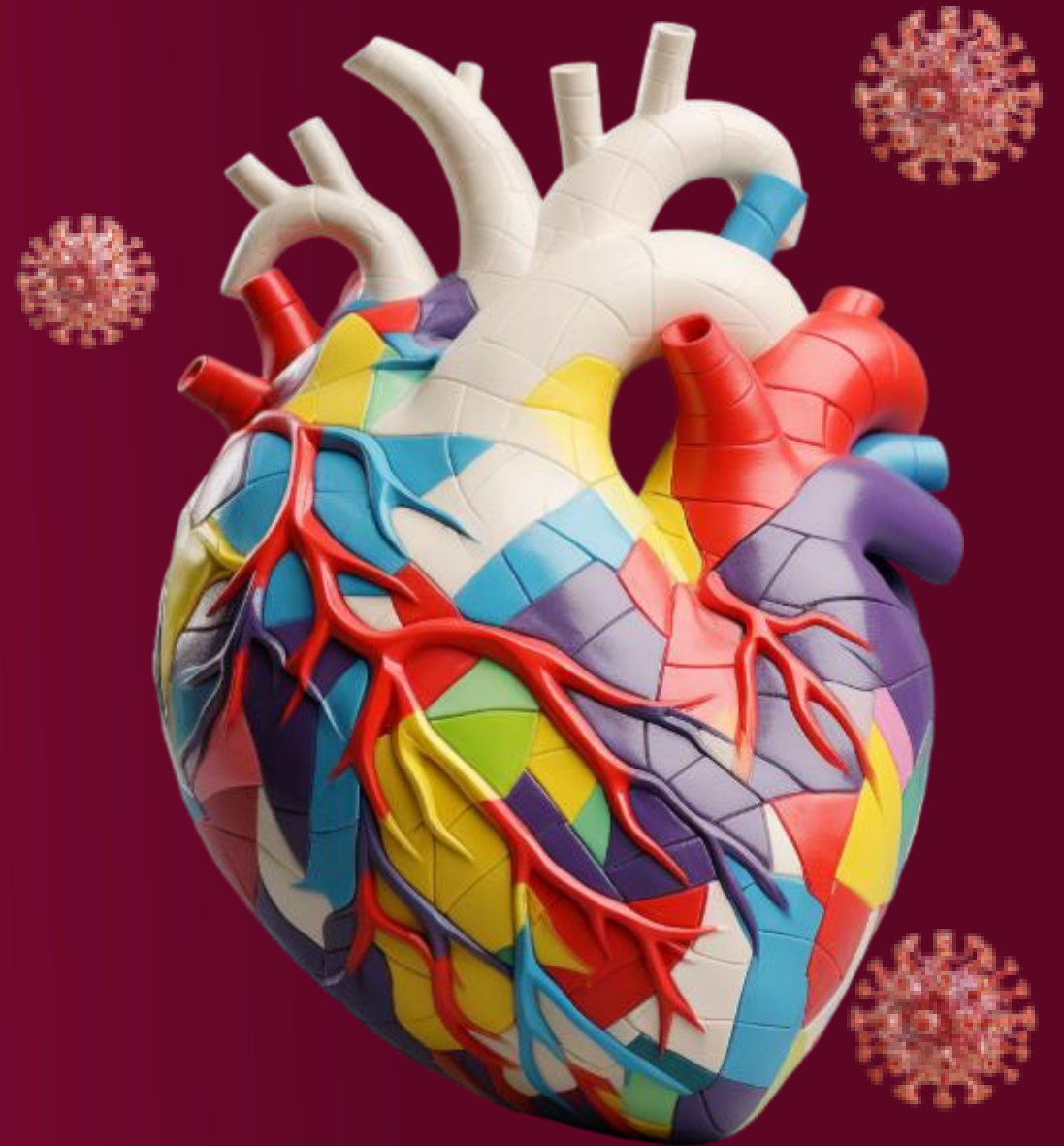


Heart Disease Prediction

Protecting what matters the most - Predict. Prevent. Protect.

Mentor – M.Gunadeep





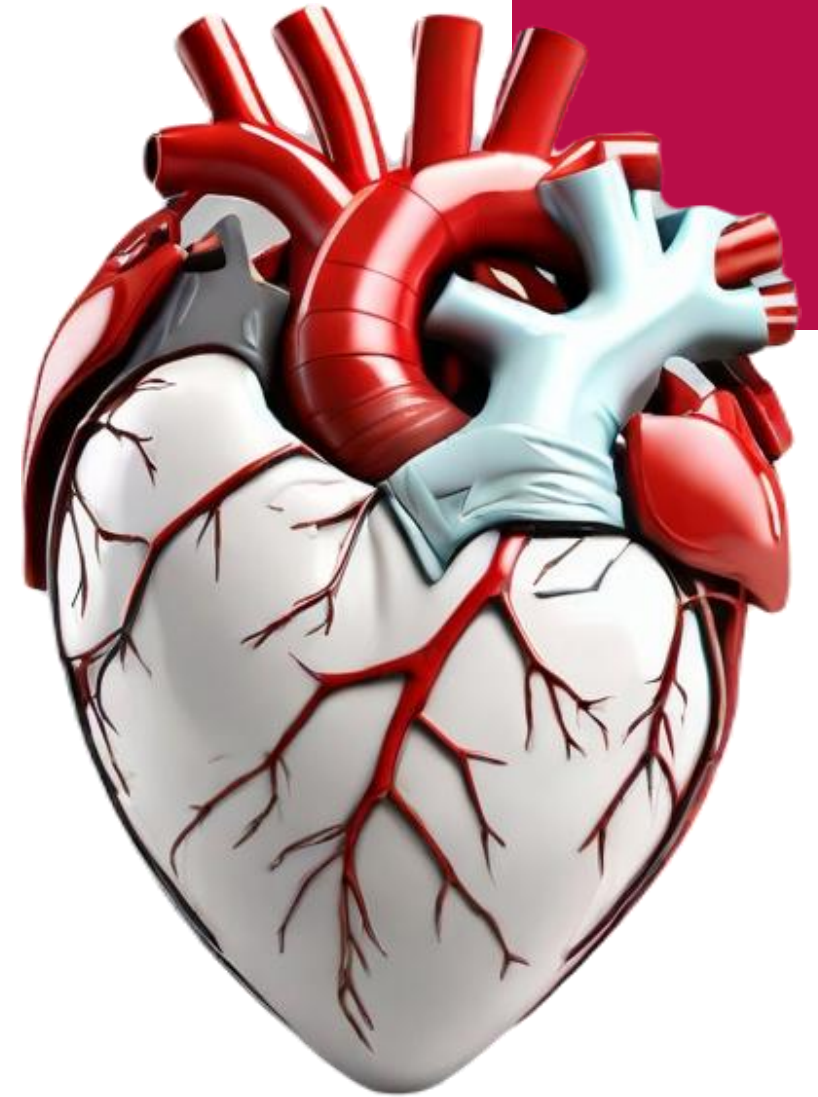
Introduction

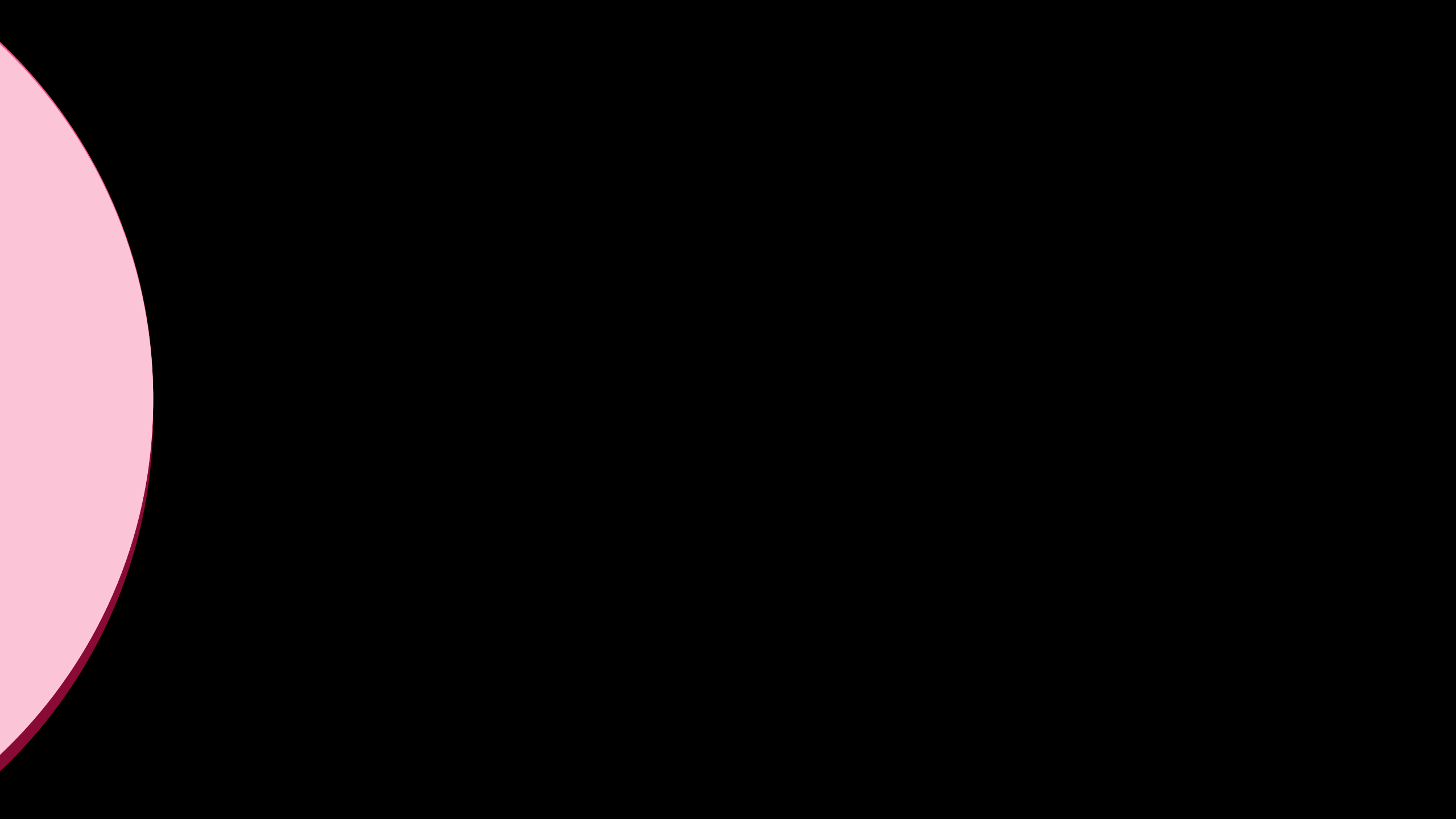
- ✓ Heart disease is a leading global cause of death
- ✓ Early detection improves prevention, intervention, and outcomes
- ✓ AI/ML models analyze health data and risk factors
- ✓ Predicts the likelihood of developing heart conditions
- ✓ Provides data-driven insights for doctors and patients
- ✓ Supports preventive healthcare and smarter decision-making



Features of Heart Disease Prediction ❤️

- ✓ Use of advanced machine learning and AI models
- ✓ Personalized risk assessment for individuals
- ✓ Continuous monitoring through real-time data
- ✓ Early identification of potential heart risks
- ✓ Support for doctors in decision-making
- ✓ Potential to reduce hospitalization and mortality rates





Tech Stack

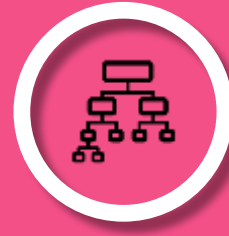


FRONTEND

HTML5 - Structure of the application.

CSS3 - Styling, responsive design, layouts for the UI.

JavaScript - Frontend interactivity, form validation.



BACKEND FRAMEWORK

PYTHON 3.X

FLASK - Building the backend API.

SQLite - Relational database for storing user data.

OCR - extract medical data from uploaded documents/images.

SQLAlchemy - That makes it easier to interact with databases in Python.



MODELS

Logistic Regression - Predicts the probability of heart disease.

Random Forest - Uses multiple decision trees to improve accuracy in classifying heart disease risk.

Support Vector Machine - Finds the optimal boundary to separate patients heart disease.

Neural Network - Learns complex patterns in heart health data to predict disease

Challenges

Handle Challenges

Challenges

- Missing and inconsistent values in dataset
- Difficulty in selecting important risk factors
- Different models gave highly varying performance results
- Unequal distribution of patients with and without heart disease
- Integration of trained ML model with the web application

Handle Challenges

- Applied proper preprocessing, handled null values, and removed duplicates.
- Using feature selection techniques to retain only impactful features.
- Compared multiple algorithms and improved with hyperparameter tuning
- Balanced dataset using resampling techniques, evaluated with ROC/recall metrics
- Connected backend model using Flask designed user-friendly UI

Future Scope

- ✓ **Smartwatches and fitness trackers:** Integration to continuously monitor heart rate, blood pressure, and ECG data.
- ✓ **cloud-based platforms:** for real-time analysis and remote doctor consultations.
- ✓ **AI-powered chatbots:** To guide patients on lifestyle changes and medication reminders.
- ✓ **Mobile applications:** For easy access to heart disease risk predictions and personalized health tips.
- ✓ **Telemedicine systems:** Expansion into for remote diagnosis and follow-up care.
- ✓ **Multi-language and region-specific apps:** To reach a wider global population.



Conclusion



- ✓ Built a system that helps predict heart disease at an early stage.
- ✓ Designed it to be simple and easy for people and doctors to use.
- ✓ Provides reliable results to support better health decisions.
- ✓ Aims to reduce risks by detecting problems before they become serious.
- ✓ Contributes to healthier lives and stronger communities.
- ✓ Developed a scalable system for future enhancements.



THANK YOU

